INCH-POUND
MIL-A-70788 (AR)
29 June 1989

MILITARY SPECIFICATION

ADAPTER, METAL PALLET

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and packaging of special adapters for metal pallets. The adapter together with the metal pallet will be used in unitizing ammunition (See 6.1).

1.2 Classification. The adapters shall be of the following types (See 6.2):

Type I	-	Adapter, Pallet, 105MM Howitzer Ammunition (Horizontal Metal Container))
Type II	-	Adapter, Pallet, 105MM Howitzer Ammunition (Vertical Metal Container))
Type III	-	Adapter, Pallet, 155MM Propelling Charge M203/M203A1 (Plastic PA133 Container)
Type IV	-	Adapter, Pallet, 155MM Propelling Charge M203 (Metal PA129 Container)
Type V	-	Adapter, Pallet, Volcano Mines M87 and M88 (PA113 Container)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140
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SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings

MILITARY

MIL-W-12332	- Welding Resistance, Spot, Seam and Production, for Fabricating Assemblies of Low Carbon Steel
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	- Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-1235	- Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1261	- Welding Procedures for Constructional Steel
MIL-STD-1660	- Design Criteria for Ammunition Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, Pa 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in affect on the date of the solicitation.

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DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

AC200000414	- Adapter, Pallet, for PA113 Container
AC200000423	- CARC Finishing Requirements for Metal Pallets and Pallet Adapters
AC200000490	- Adapter, Pallet, 105MM Howitzer Metal Container (Horizontal)
AC200000505	- Adapter, Pallet, 155MM Propelling Charge (Plastic Container)
AC200000506	- Adapter, Pallet, 155MM Propelling Charge (Metal Container)
AC200000524	- Adapter, Pallet, 105MM Howitzer Metal Container (Vertical)

(Copies of specifications, standards, drawings and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity of as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951	- Standard Practice for Commercial Packaging DoD Adopted

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Adapters. The adapter assemblies shall comply with all requirements specified on drawings AC200000414, AC200000490, AC200000505, AC200000506 and AC200000524, as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Welding. The welding shall comply with the requirements of MIL-W-12332 or MIL-STD-1261 and weld callouts in the applicable drawings.

3.4 Lifting ring pull test. Each lifting ring of the adapter top assembly shall withstand a (non-destructive) load of 2,500 pounds minimum (min.)

3.5 Painted surface. The painted surfaces shall comply with 3.5.1 or 3.5.2, as applicable.

3.5.1 Conventional coating. The painted surface of the adapters shall comply with the requirements of the applicable drawings, TT-C-490 and with instructions in MIL-STD-171.

3.5.2 Chemical agent resistant coating (CARC). The painted surfaces of the adapters shall comply with the requirements of dwg AC200000423 and applicable drawings.

3.6 Unit load tests. The metal pallet adapters, as part of unit loaded pallets, shall conform to the requirements specified in MIL-STD-1660 (see 4.5.2).

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship.

3.8.1 Adapters. The adapters shall be regular, smooth, and free from wrinkles, cracks, rough spots, burrs, sharp edges, and any other defect that might affect the serviceability, durability and safety of the adapters. Weld splatter is permissible on non load bearing surfaces provided it is firmly attached.

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3.8.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided they are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint chips which expose primer or bare metal are acceptable provided they are not more than 0.25 inches in diameter and no more than two such paint chips on an adapter. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inches in length and no more than 10 inches total on an adapter. Bare metal on contact surfaces in areas inaccessible to paint is permissible.

3.8.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (See 4.3)

b. Quality conformance inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part description-	Quantity
Top assy, painted	12
Bottom assy, painted	12
Adapter assy	12

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-A-70788 (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Assemblies and Components		SHEET 1 OF 1		As applicable
	Top Assembly, Painted				
	Examination for defects	12		3.2	4.4.2.1
	Lifting ring pull test	6		3.4	4.5.1
	Bottom Assembly, Painted				
	Examination for defects	12		3.2	4.4.2.2
	Adapter Assembly				
	a. Examination for defects	12		3.2	4.4.2.3
	b. Unit load tests	3		3.6	4.5.2

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.2 Examination. Inspection for critical defects (and major defects when so specified) shall be 100 percent. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70788 (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.1	Top Assembly		1	1	NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lifting ring pull test		0.65%	3.4	4.5.1
102	Overall length		1.0%	3.2	Gage
103	Overal width		1.0%	3.2	Gage
104	Protective coating thickness (total thickness minus primer thickness (see dwg AC200000423 and Notes 1 & 2 below))		1.0%	3.5	Gage
105	Weld missing or incomplete		1.0%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.5%	3.8	Visual

NOTES:
1. This inspection only applies when the assembly has been painted with CARC.
2. The primer thickness shall be determined from a test panel.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER AC200000414, 491, 505, 506 or 525 as applicable
4.4.2.2	Bottom Assembly				NEXT HIGHER ASSEMBLY AC200000414, 490,524, 505 or 506 as applicable
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		1.0%	3.2	Gage
102	Overall width		1.0%	3.2	Gage
103	Protective coating thickness (total thickness minus primer thickness (see dwg AC200000423 and Notes 1 & 2 below))		1.0%	3.5	Gage
104	Weld missing or incomplete		1.0%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.5%	3.8	Visual

NOTES: 1. This inspection only applies when the assembly has been painted with CARC.
2. The primer thickness shall be determined from a test panel.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70788 (AR)

PARAGRAPH	TITLE		SHEET OF	DRAWING NUMBER
4.4.2.3	Adapter Assembly		1 1	AC200000414, 490, 524, 505, 506 as applicable
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Part missing or incorrectly applied		1.0%	3.2	Visual
Minor					
201	Protective finish missing, incomplete or incorrectly applied		1.5%	3.8.2	Visual
202	Marking missing or incorrect		1.5%	3.2	Visual
203	Evidence of poor workmanship (other than protective finish)		1.5%	3.8	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1 and in 4.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. When Automated Acceptance Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection. (See 6.4)

4.5.1 Lifting ring pull test. Each lifting ring of the adapter top assembly shall be placed in a test machine designed for pull testing. The load shall be applied by a steady pressure and the load read directly by a calibrated gage. The load applied to each ring shall be 2,500 pounds. The results of the test shall not show signs of deformation, change in length of ring nor any separation of ring from top assembly.

4.5.2 Unit load test. (First article inspection only) Prototype unit loads for the pallet and adapter assemblies shall be fabricated in accordance with MIL-STD-1660 with the following exceptions. The gross weight of the ammunition load shall be 2,700 pounds minus 50 pounds, evenly distributed over the deck and shall obtain a height of 48 inches $\pm$ 2 inches to simulate the configuration of the final unitized load. Fabrication of the unit load shall include using the exact container for which the adapter is designed or a suitable facsimile. The following tests, on the loaded pallets, shall be performed in accordance with MIL-STD-1660.

4.5.2.1 Repetitive shock test. The acceptance/rejection criteria specified in section 7 of method 5019.1 in FED-STD-101 is mandatory.

4.5.2.2 Edgewise drop (rotational) test. The acceptance/rejection criteria specified in section 7 of Method 5008.1 in FED-STD-101 is mandatory.

4.5.2.3 Incline impact test. The acceptance/rejection criteria specified in section 7 of Method 5023 in FED-STD-101 is mandatory.

4.5.2.4 Sling compatibility test. The acceptance/rejection criteria specified in section 5 of MIL-STD-1660 shall apply.

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5 PACKAGING

5.1 Preservation and packing. None required.

5.2 Packing.

5.2.1 Commercial. The adapters shall be packed in accordance with ASTM-D3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM-D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the adapters. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If adapters are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When adapters from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

6.1 Intended use. These metal adapters are intended for use with the metal pallets when unitizing packaged ammunition containers.

6.2 Acquisition requirements. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

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6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the acuracy and precision and drawings of any special equipment required.

6.5 Subject term (key word) listing.

Adapter
Ammunition
Container
Metal
Pallet
Unitizing

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 8140-A776)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70788 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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